Leak-detection model: what the pilot can and cannot tell us

Postbrief from the 14 March model-review call (project lead, data scientist, field engineer, analyst). Illustrative example; draft for discussion.

Why this matters. The utility wants to send repair crews where leaks are likely. The chain runs pressure sensors → the leak model → a ranked list of pipe segments → a crew dispatch. Two questions decide the next quarter: *is the ranking good enough to dispatch on?* and *which fix improves it most: more sensors, better pipe records, or a better model?*

1. Words we use, and when to ask for each

Term	Plain meaning	Ask for it when…
Sensitivity (feature importance, local or global)	How much the model's score moves when one input moves. Says nothing about how wrong that input is.	"Does the model even react to pipe age?" If not, better age records can never help.
Uncertainty (predictive uncertainty, calibrated intervals)	How sure the model is about each segment's score.	"Can we tell a crew chief this segment is 80% likely to be leaking?"
Accuracy (validation: precision and recall against confirmed leaks)	How often the ranking matches what crews actually found. Needs field outcomes.	"Of last month's top 50 segments, how many really leaked?"
Error attribution (ablation, data-swap tests)	Which input or assumption causes the misses.	"We missed the Elm St. leak. Was it a sensor gap or a wrong pipe material?"
Value of new data (information gain, data-acquisition experiment)	How much accuracy improves if we add a data source.	"Are 40 more sensors worth $200K, or would cleaner pipe records do more?"
Label noise (unreliable ground truth)	Confirmed-leak records that are themselves wrong or late. Inflates or hides errors.	"Before we trust the recall number, how often are work orders mis-coded?"

Rules of thumb. Importance ≠ value: a feature the model leans on can already be well measured. Confidence ≠ accuracy: a sure model can still be wrong, and only field outcomes show it. The lead's question, "what would most improve the dispatch list?", is **value of new data**, which needs accuracy first.

2. The proposed plan

Does: retrains the model on 18 months of sensor data and ranks features by importance (SHAP values), about two weeks. **Will measure:** which inputs the model leans on; a cleaner, faster pipeline. **Will not measure:** accuracy against confirmed leaks; the effect of new sensors; label noise. **Against the need:** about a quarter of the way. It shows what the model uses, not what would make the list better, which is the lead's real question: "if we buy sensors and nothing changes, I've spent the budget."

3. Quick ways forward

Add-on (technique)	What it gives	Workdays
A. Score last quarter's list (retrospective validation: precision@50 against work orders)	The first honest accuracy number	2
B. Audit 100 work orders (label-noise estimate)	Tells us how far to trust A	2–3
C. Hold out sensors (data-denial experiment: drop 20% of sensors, re-score)	Previews what adding sensors would buy, before buying them	3
D. Swap in corrected pipe records for one district (data-swap ablation)	Compares the records fix against the sensors fix	3–4

Shortest useful path: A, then C and D side by side, in about two weeks. That turns "buy sensors?" into a measured comparison.